

Linkedcare Dental Clinic SAAS Platform

#Dental #Clinic Management #SAAS #User Research #User Testing

DENTAL CLINIC SAAS PLATFORM

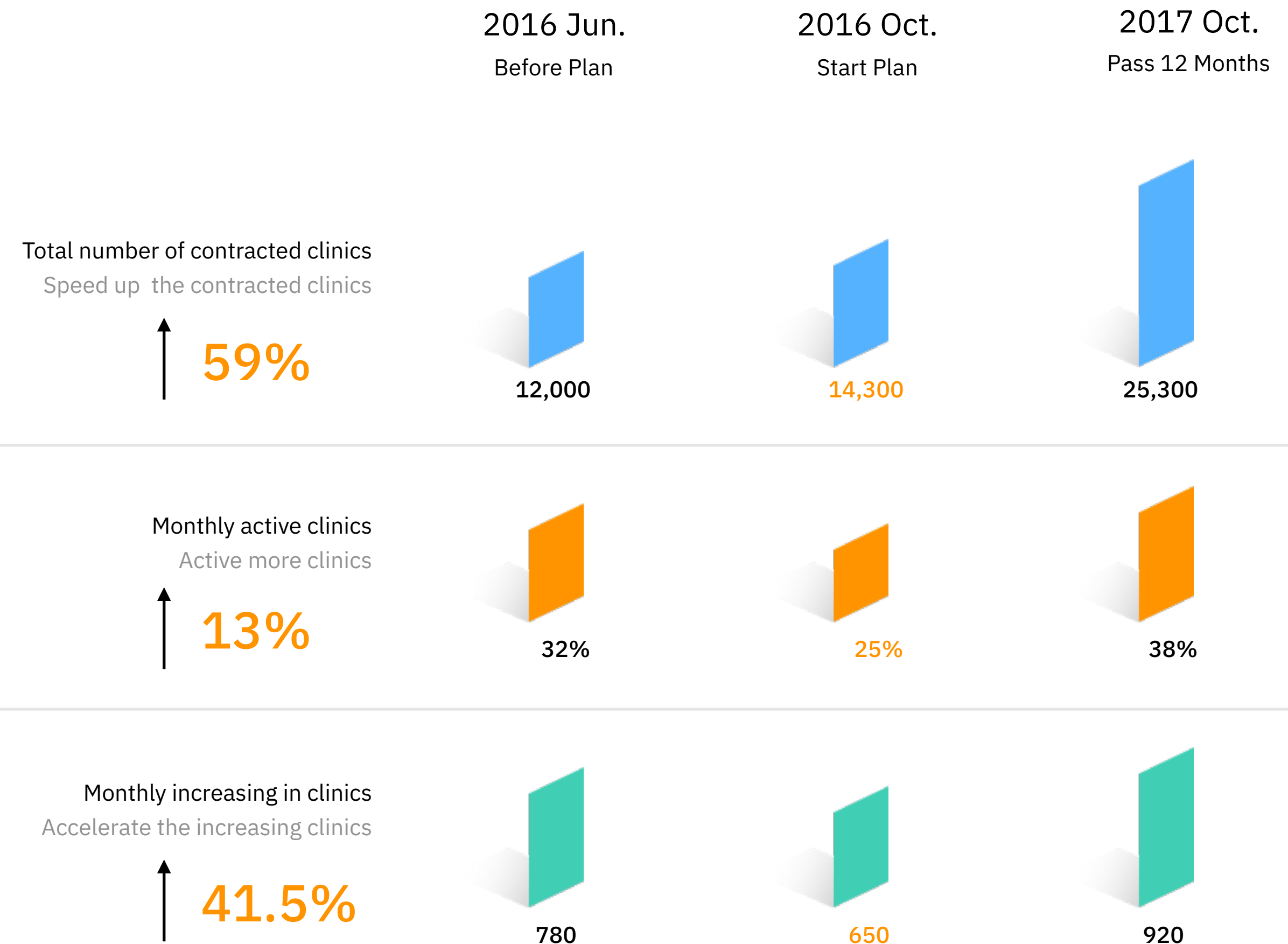
In dental clinic SAAS platform has clearly definition of authority management. Key features of this platform are include the ability to cover administration, such as a calendar and a way to schedule, track, and manage appointments. The billing section records fees and automate claims with national health insurance, as well as track payment status and automate billing reminders as required. This platform also integrate with imaging equipment and x-ray sensors to make it easier to upload to platform.

There is also integrate a APP for doctor to help them review EMR on mobile. However, the APP has the ability to share treatment information and diagnosis history with patients.

RESPONSIBILITIES

User Interview
Customer Journey Map
Prototype & UX/UI Design

The Outcomes of User Experience Improvement Plan





USER RESEARCH PLAN

PLAN

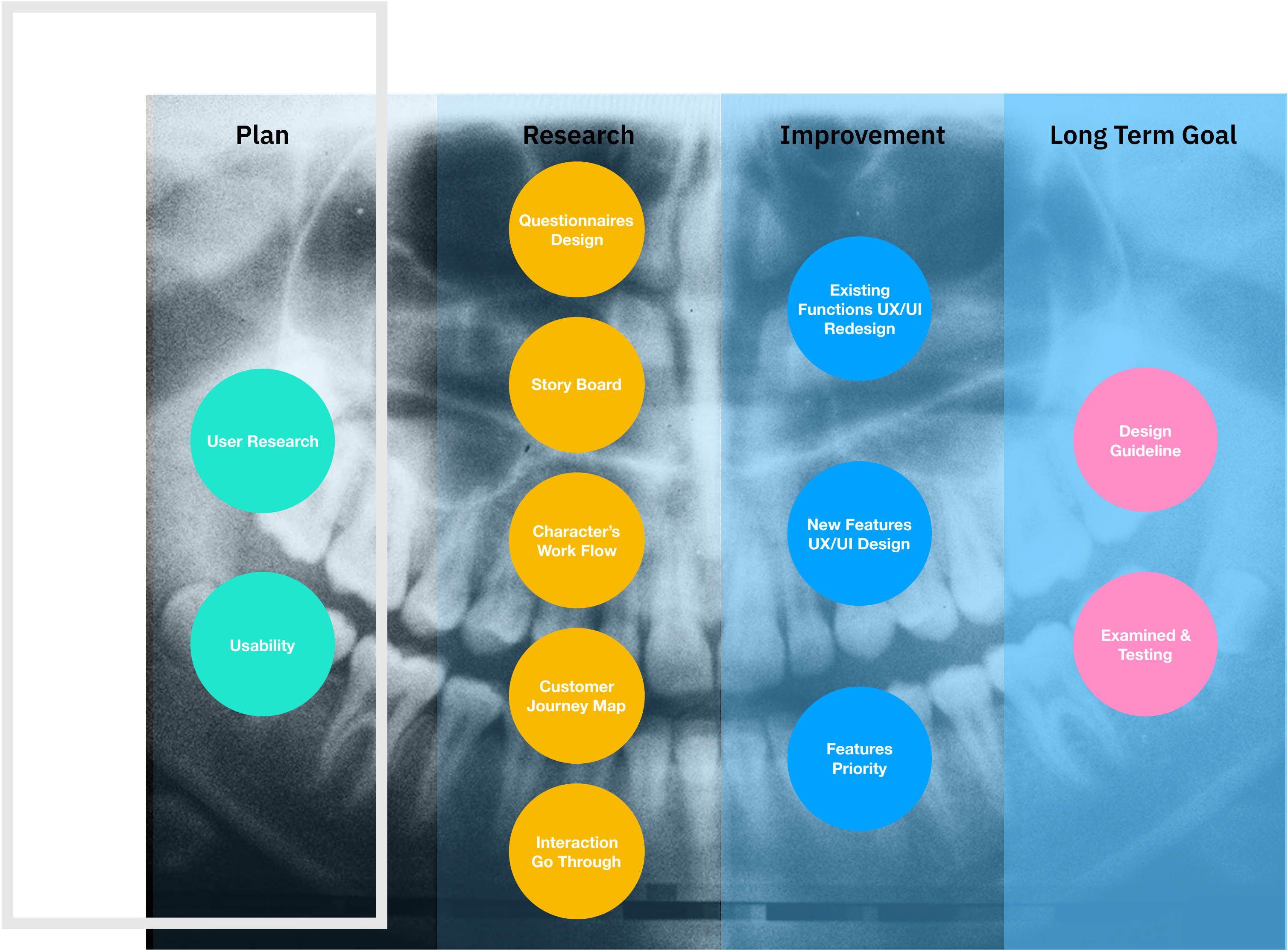
First of all, we need to figure out which functions are user needs, improve existing functions of UX and UI at the same time. So I made up a user research plan to improve the usability of existing SAAS platform. Also need to solve the biggest problem of continuously losing of contracted clinics.

USER RESEARCH

We need to clarify which function are real user needs, so set up two actions to help us to know. Character’s workflow and CJM both are good way to understand our user’s behavior

USABILITY

We designed questionnaires and review the UX/UI to identify what’s problem on our platform. So we can prioritize the issues synced with PM and engineers.





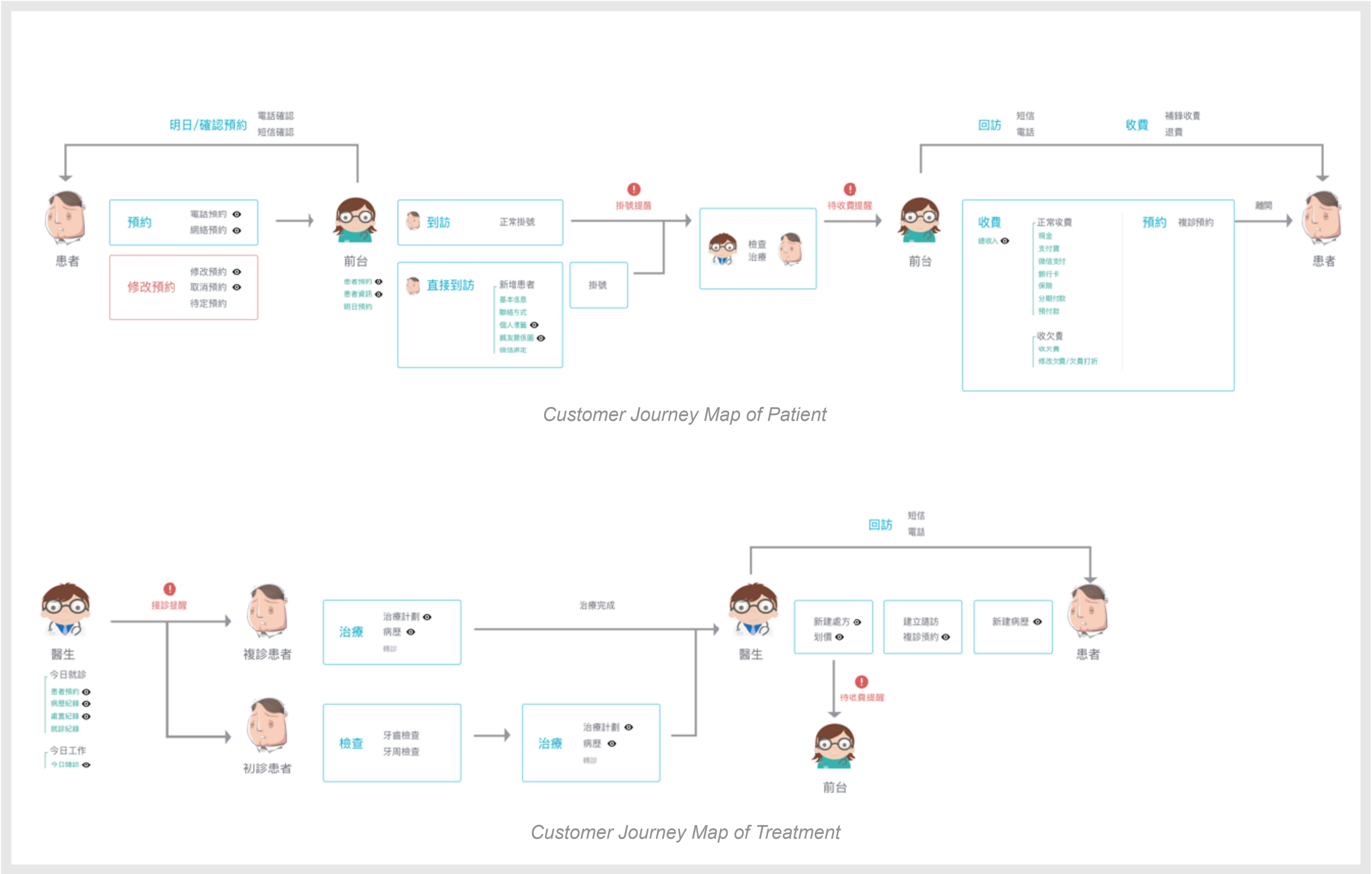
USER RESEARCH PLAN

STORYBOARD

We designed the questionnaires for each character and visualize the storyboard for two perspective of patient and clinic. We also observed user's behavior after interview.



Questionnaires list



Customer Journey Map of Patient

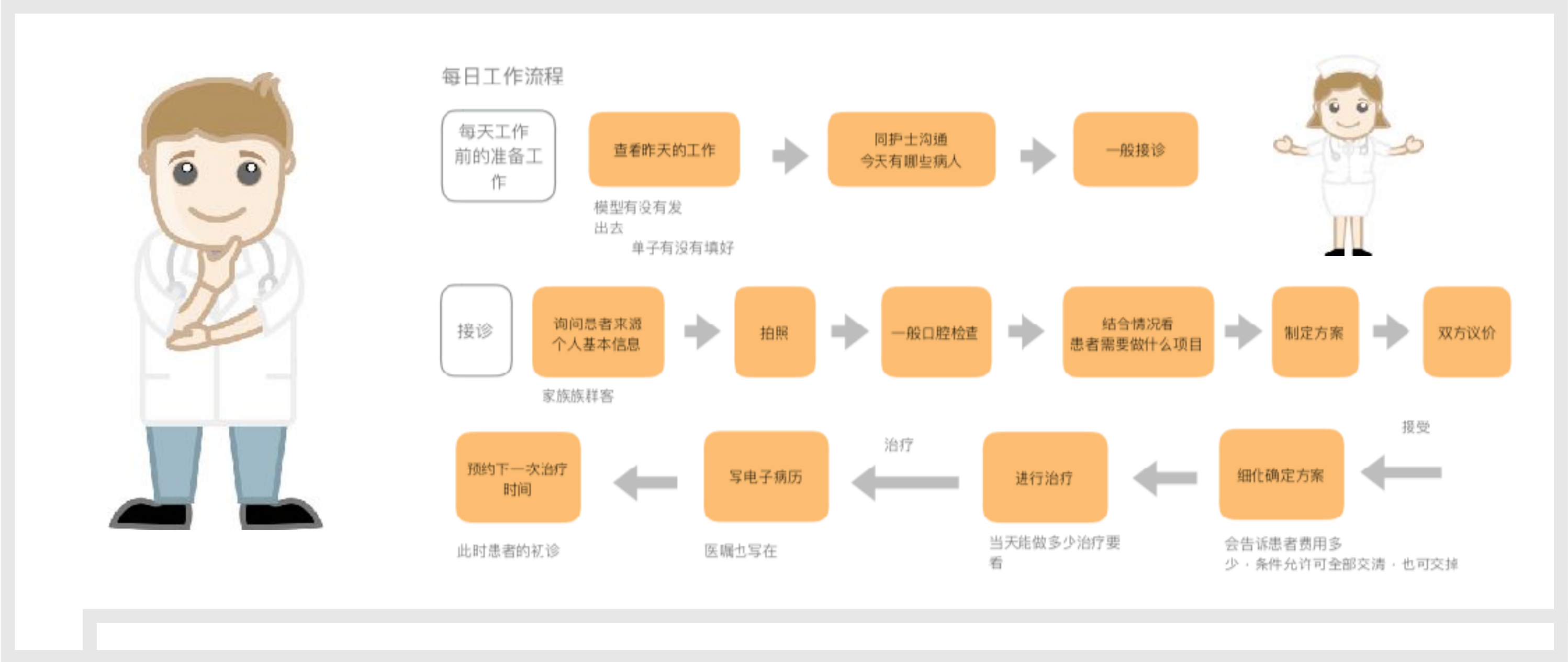
Customer Journey Map of Treatment



USER RESEARCH PLAN

DOCTOR'S WORK FLOW (Nurse included)

For doctor, we recorded their daily job and how do doctor treat the patient in dental clinic. Doctor in dental clinic is responsible to record the EMR(electronic medical record) and provide the solution for patient. Nurse were assigned to doctor and settle down all preparation work before patient arrival.



RECEPTIONIST'S WORK FLOW

Receptionist always take the most heavy works in dental clinic, including booking, record old and new patient's basic information, maintain the EMR, be the cashier. In our research we found out, receptionist also play the most important role, recall and patient care after treatment.





USER RESEARCH PLAN

RECEPTIONIST’S WORK FLOW

Scenario 1

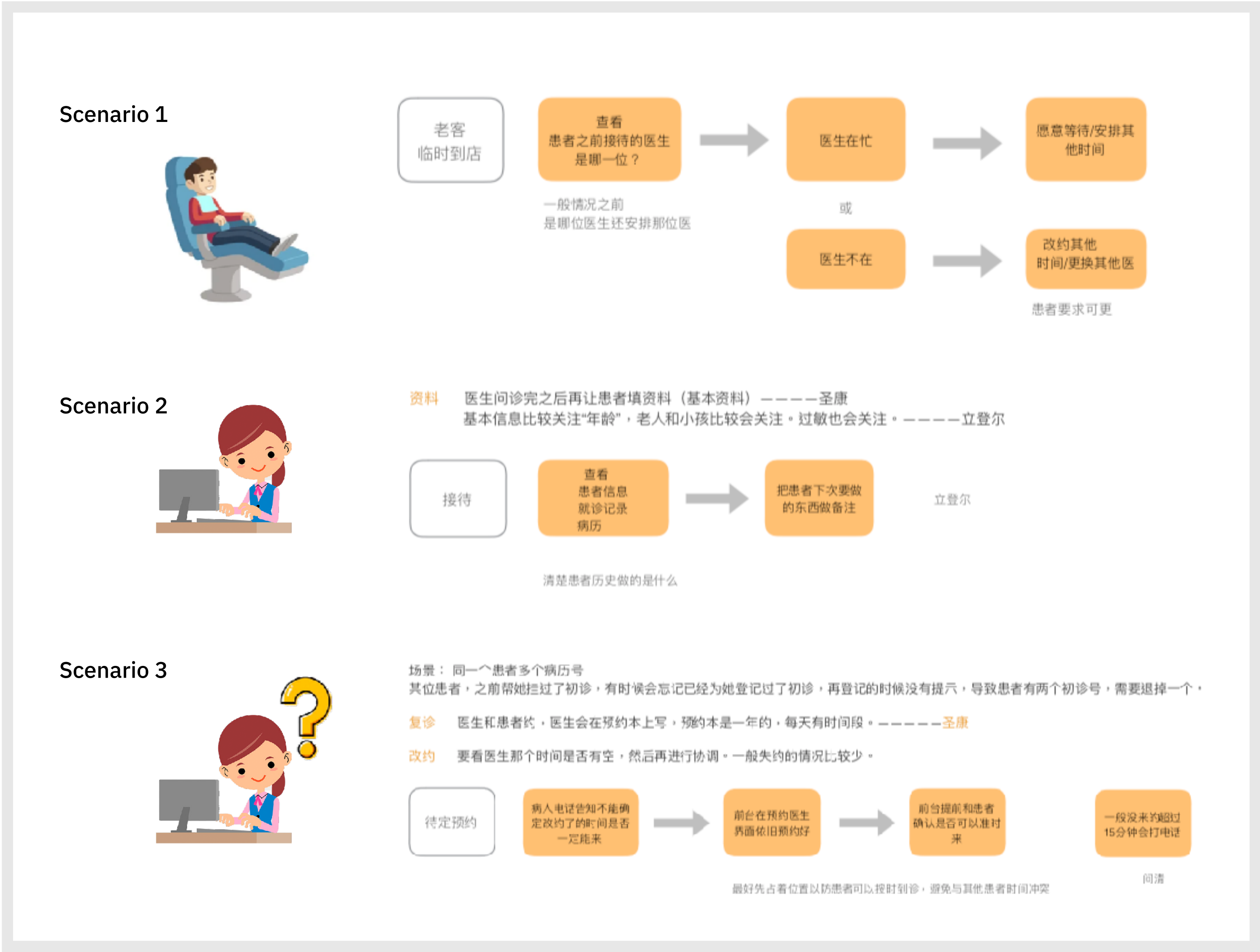
Regular customers arrived at clinic without booking, so receptionist need to ask if the patient is willing to wait or book another time on platform. If emergency, receptionist will ask doctor’s available time and schedule for patient, like a consulting role.

Scenario 2

We tried to find out what information or touch points while schedule the receptionist care about and what characteristic (like age) of the patient they more concern. What different of before and after treatment they be to pay close attention to ?

Scenario 3

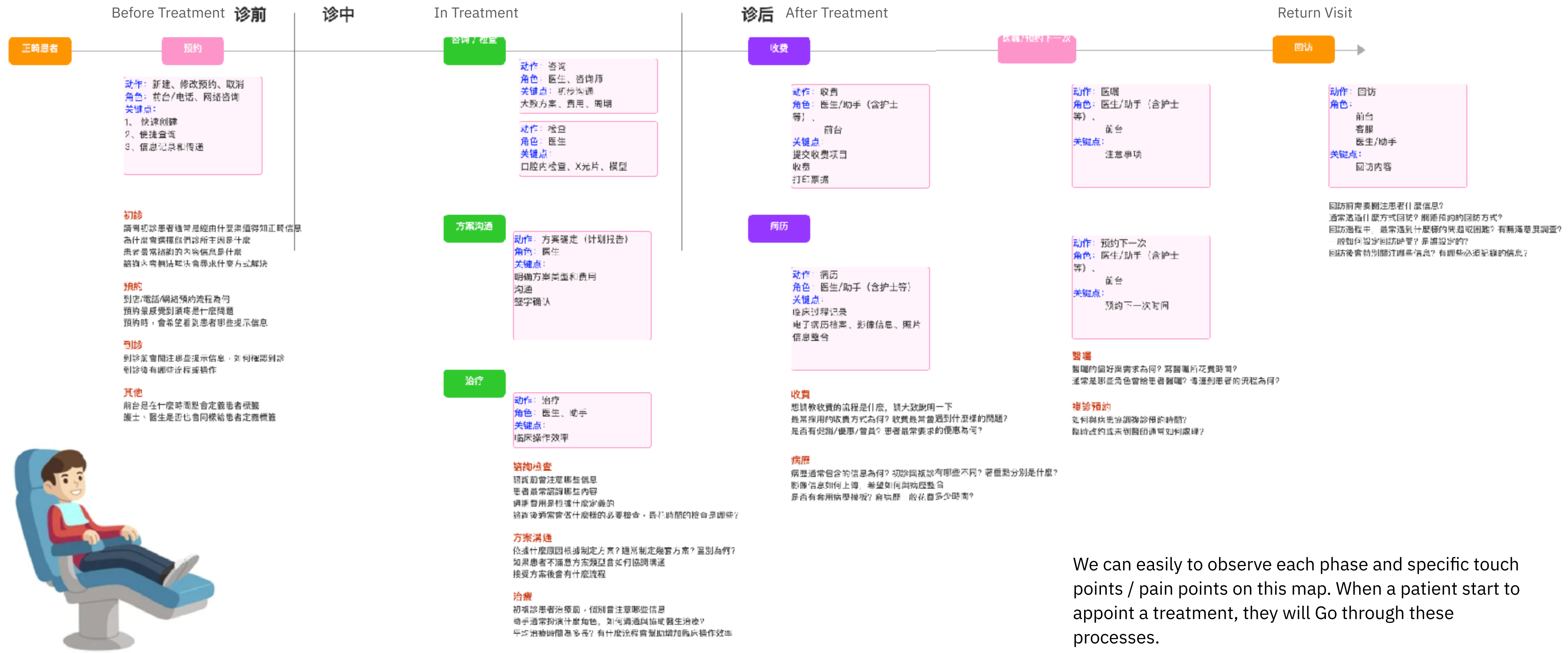
A patient may have different account in our system (can not be search, so create a new account is allowed in our system) because of relationship of their family. So we need to fix this bug and solve the problem in this scenario.





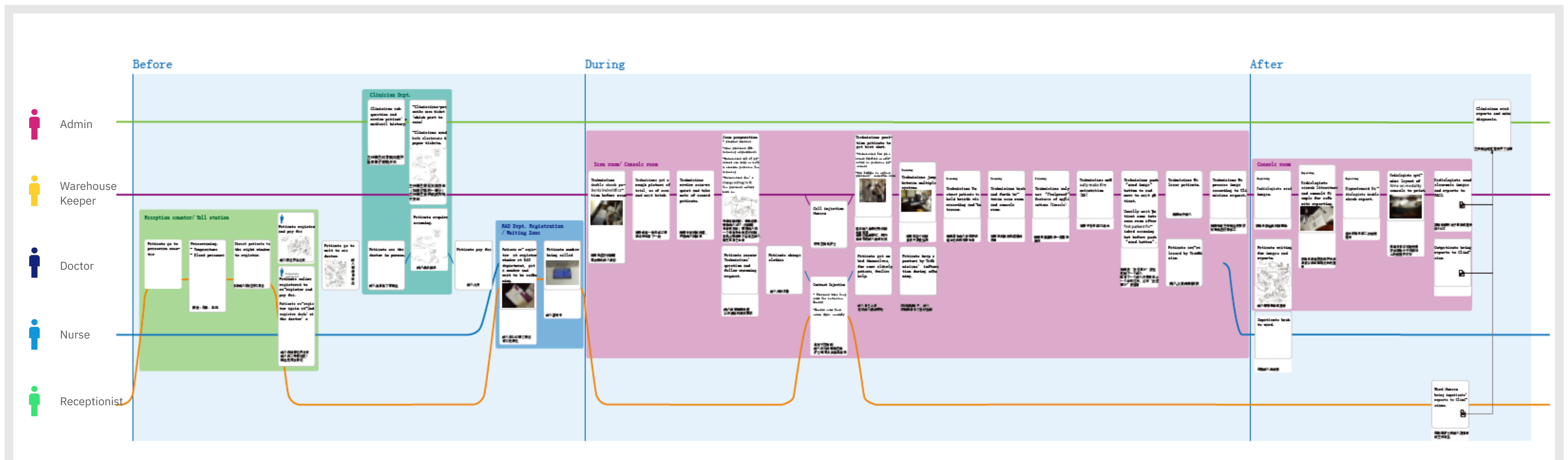
USER RESEARCH PLAN

TOUCH POINTS IN TREATMENT PROCESS



We can easily to observe each phase and specific touch points / pain points on this map. When a patient start to appoint a treatment, they will Go through these processes.

First we need familiar with our customers, accompany with them about 3 or 4 days and observe the touch points through their daily work. After finished all of interviews, questionnaires, behavior observation of user, our team summed up these path. Combined with issues from dental clinic, PM and pre-sale with designers designed user journey map for each character on our platform.

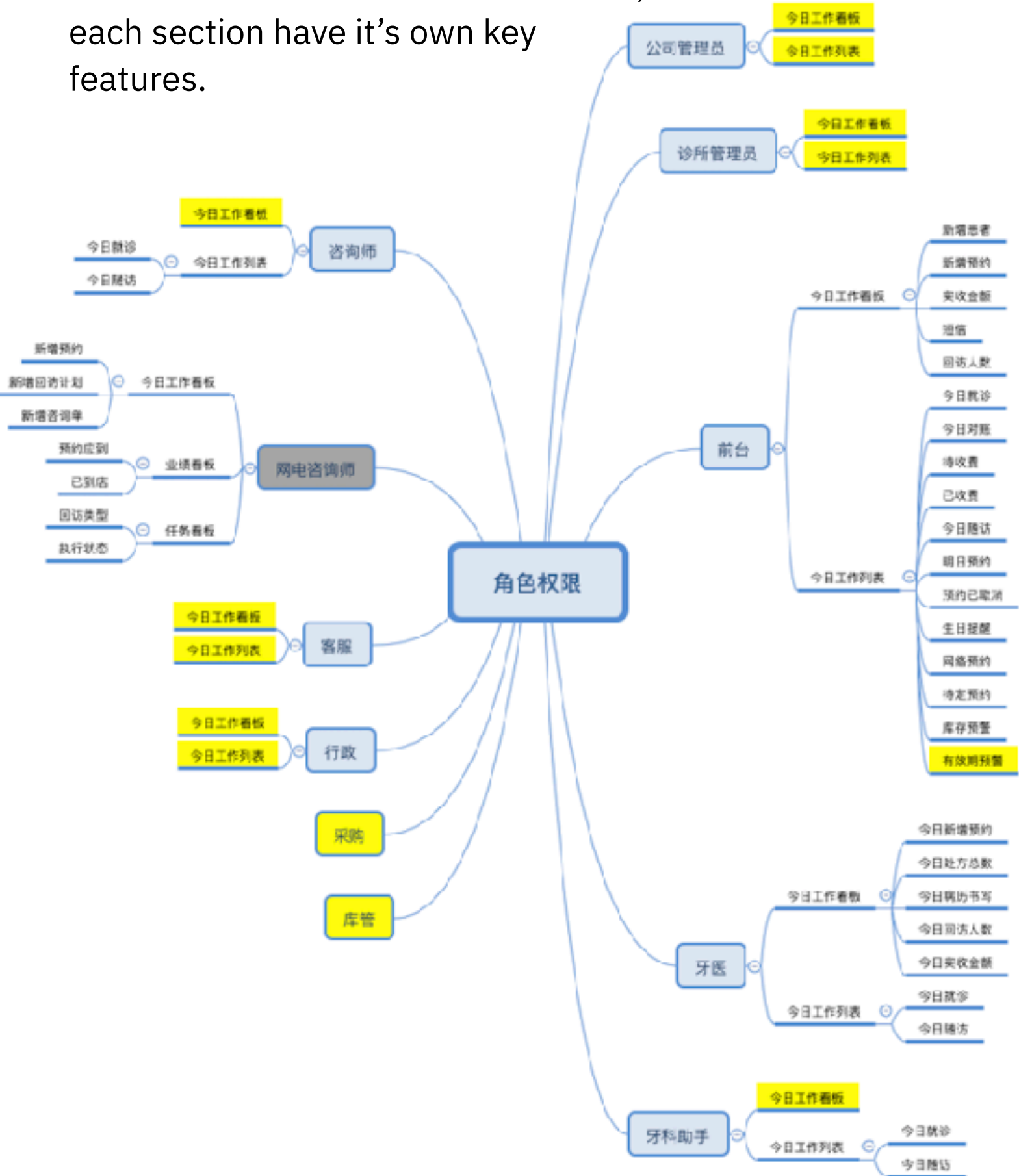




IMPROVMENT

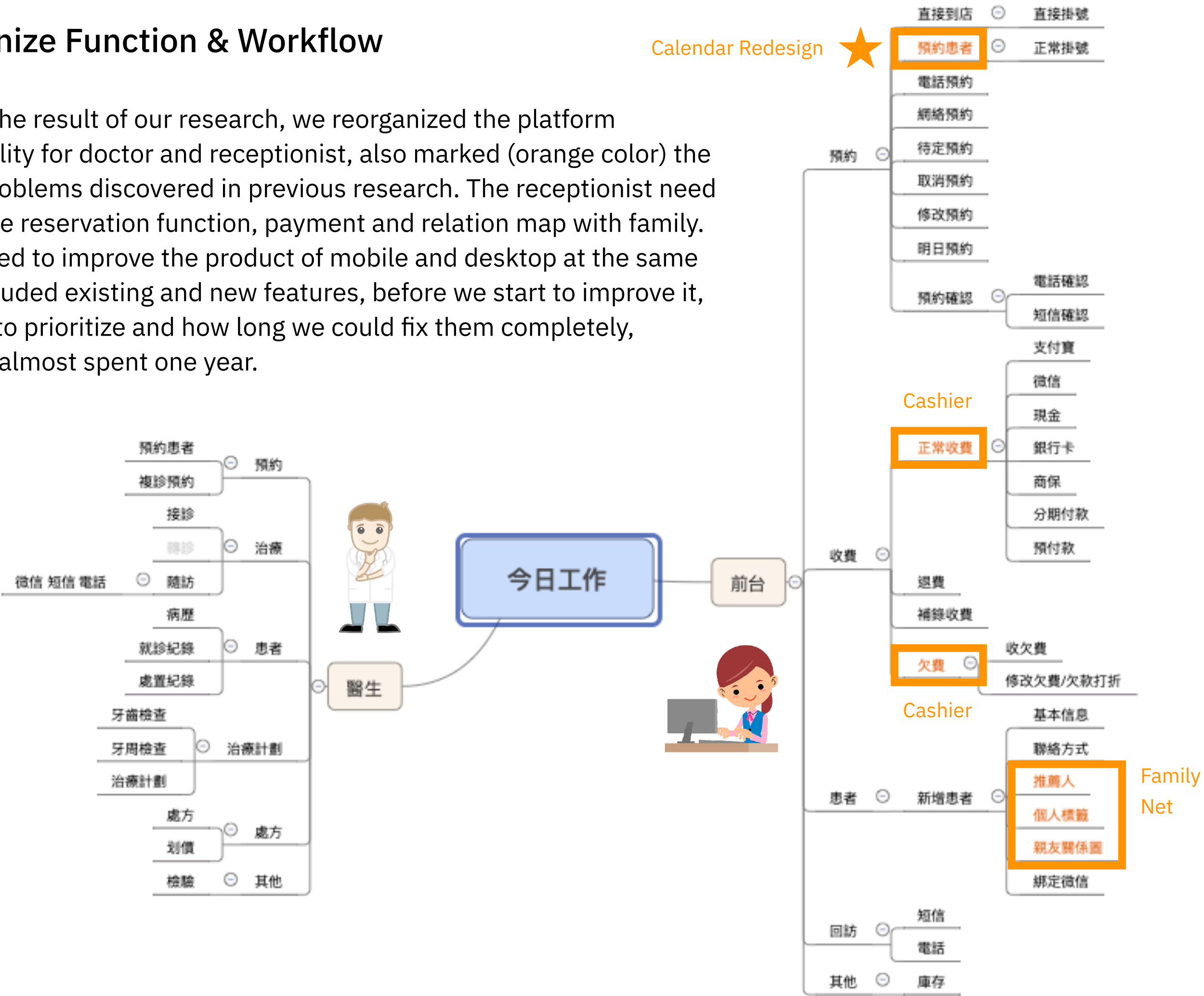
Role Authorization

We clarified each role in dental clinic, each section have it's own key features.



Reorganize Function & Workflow

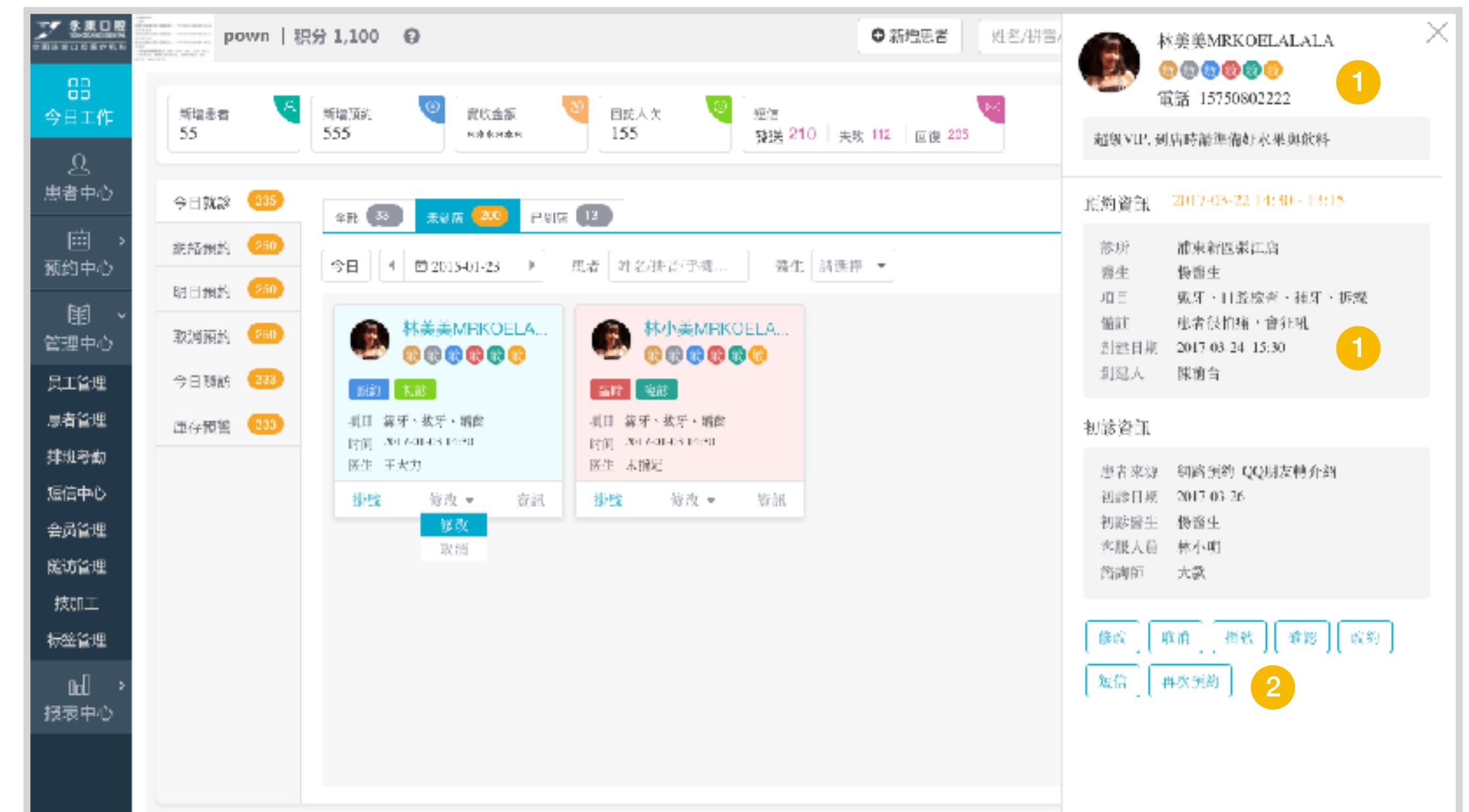
Base on the result of our research, we reorganized the platform functionality for doctor and receptionist, also marked (orange color) the import problems discovered in previous research. The receptionist need to improve reservation function, payment and relation map with family. We decided to improve the product of mobile and desktop at the same time. Included existing and new features, before we start to improve it, we need to prioritize and how long we could fix them completely, although almost spent one year.



IMPROVMENT



1. Enhance category and status visualization
2. Customize colors base on patient's identification
3. More status icon display for patient's card
4. Add status of treatment office



1. Real-time to see patient's basic info & appointment history
2. Add tag feature to quick recognize extra information



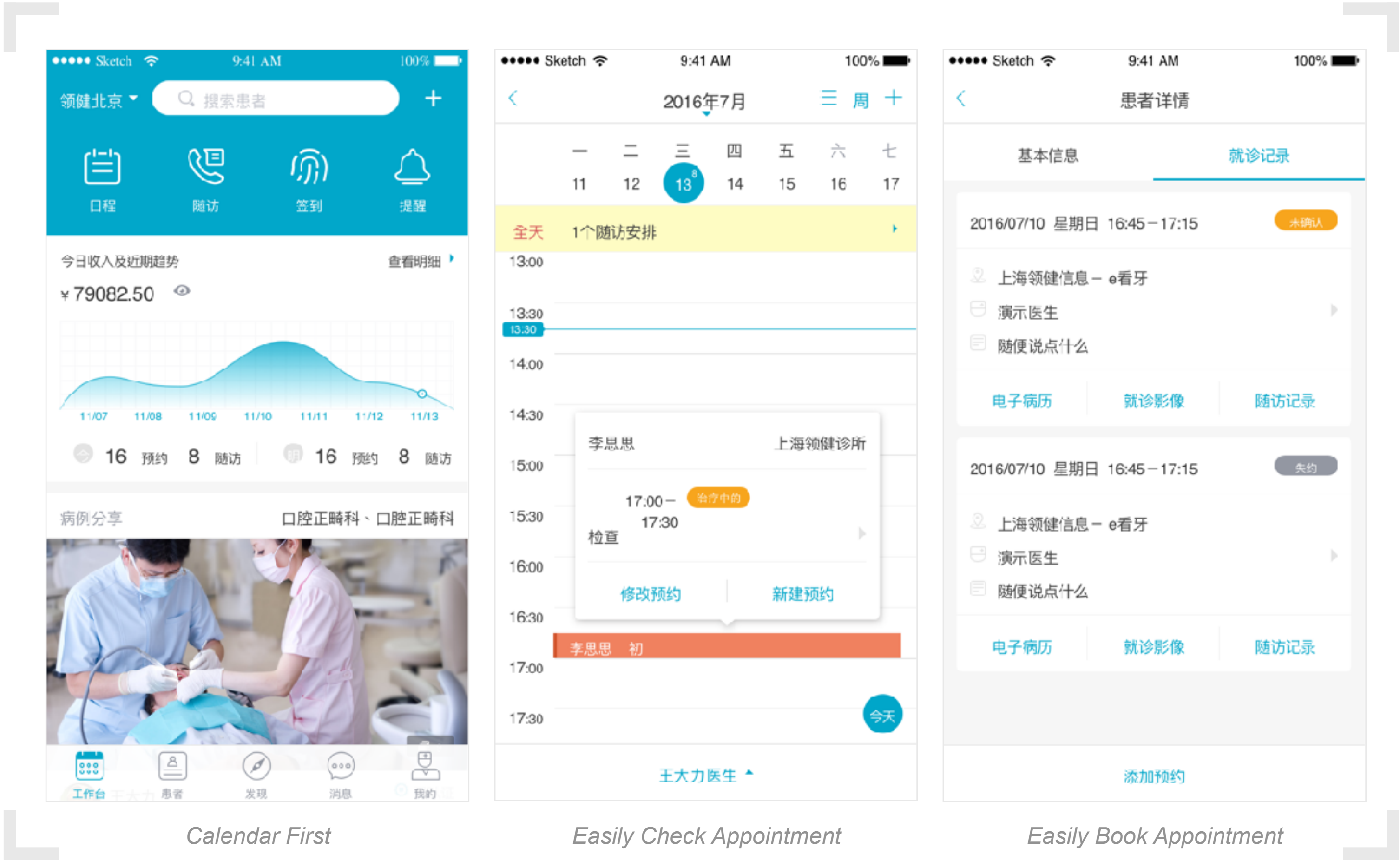
IMPROVMENT

CALENDAR REDESIGN

Base on summary of final CJM, we designed the prototypes to confirm with our sales and pre-sales. To make sure our prototypes are fits user needs. After confirmed with them, we started to redesign UX / UI and deliver these pages (include desktop and mobile) before next launch. Each launch cost us two or three weeks, we regularly confirmed with our client every two launch.



Prototypes of Calendar



Calendar First

Easily Check Appointment

Easily Book Appointment

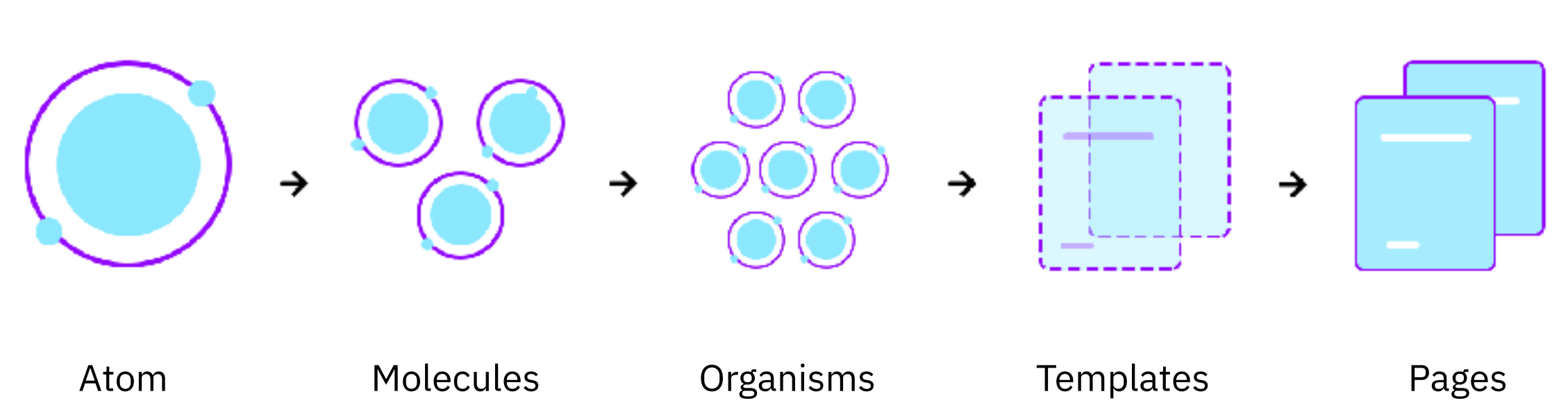


IMPROVMENT

DESIGN GUIDELINE

From Atoms to Entire Products

Atomic design borrows from that way of looking at things, applying it to our products. From UI components to maintain UX consistency. In Atomic design theory, the components are atoms, the form is the molecule. The entire page is a living organism.



Basic Setting

Font

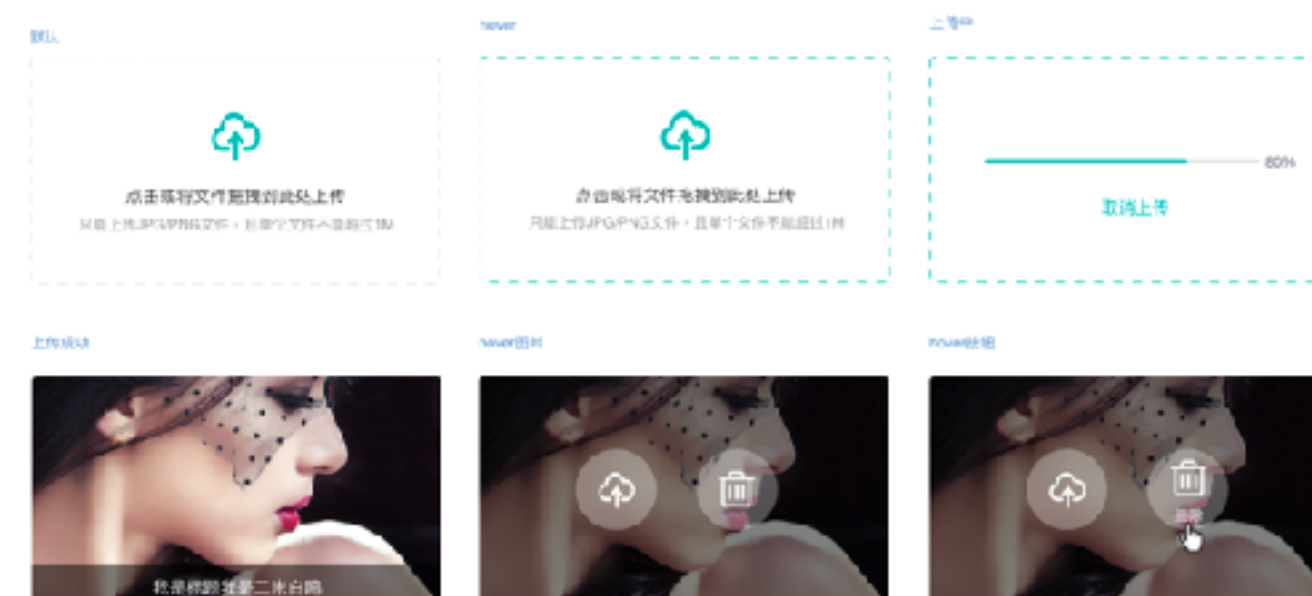
中文字体	英文/数字字体	字体大小
我是字体 Source Han Sans CN	English520 Helvetica Neue	我是页面标题 18px - Medium
		我是主标题 16px - Medium
		我是次标题 14px - Medium
		我是正文 14px - Regular 建议行高20px
		我是小标题 12px - Medium
		我是辅助文字/正文 (小) 12px - Normal 建议行高18px
		超过18px大小的字体根据实际情况自定义

Color

字体颜色	背景/边框颜色	特殊颜色	其他颜色 (表格等)
#2F2E2E 强调/重要文字	#C3D3D3	#FF4949	#26A4D6
#575D6A 内容/辅助性文字	#666A73	#FFCDB8	#D9F4FF
#9397A2 提示性文字/失效文字	#9397A2	#20A0FF	#5A8ECA
#197ED0 链接文字	#E3E3E3 正常边框/线	#13CE66	#7A84C1
#D3DCE6 深色背景文字	#EBEBEB 禁止	#D3FCE5	#F0F2FF
#FFFFFF 深色背景文字	#F5F5F5	#F7BA2A	#FAEDFF
	#FAFAFA 全域背景色	#FDF1D4	#A99F19
			#D161A1
			#FE8F5
			#94EF3C
			#EAF2D8
			#E2534D
			#FE0ED
			#27A393
			#D4EDE9
			#3BECEF
			#D8F2F2

Components

Layout



Navigation





Mobile Guideline - Basic Layout Design



Login

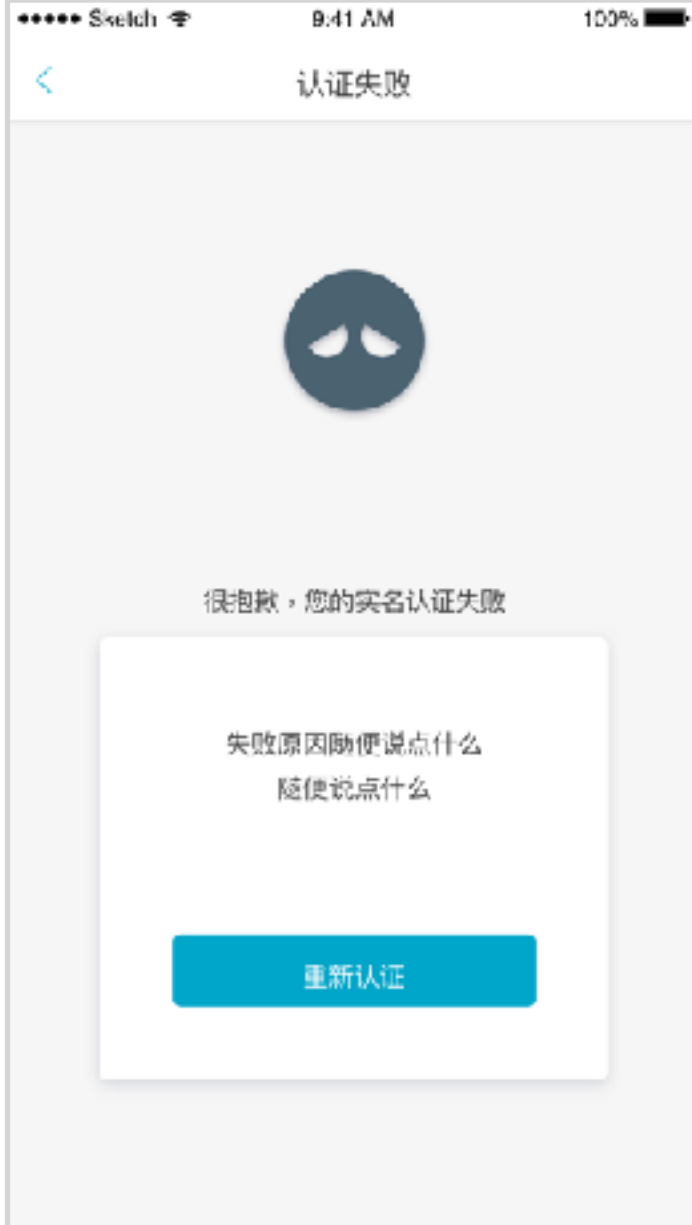
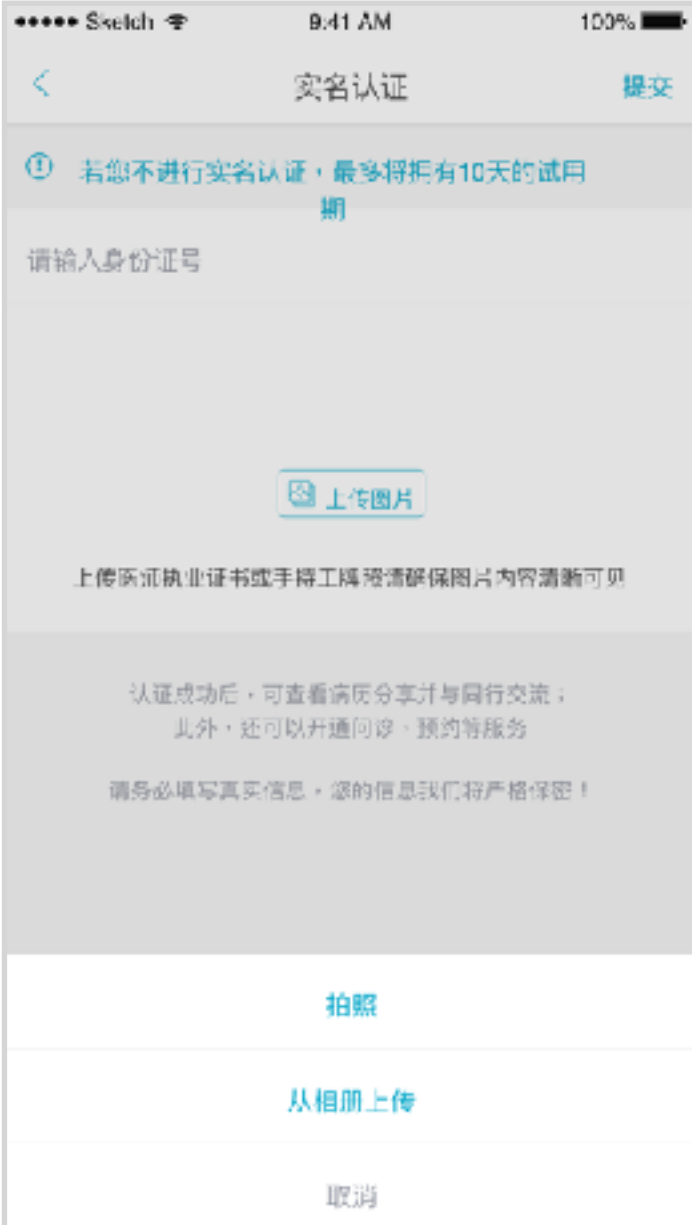
List

Upload

Reacting Button

Drop Down Button

Pop-up Reminder





EXAMINED & USERS TESTING

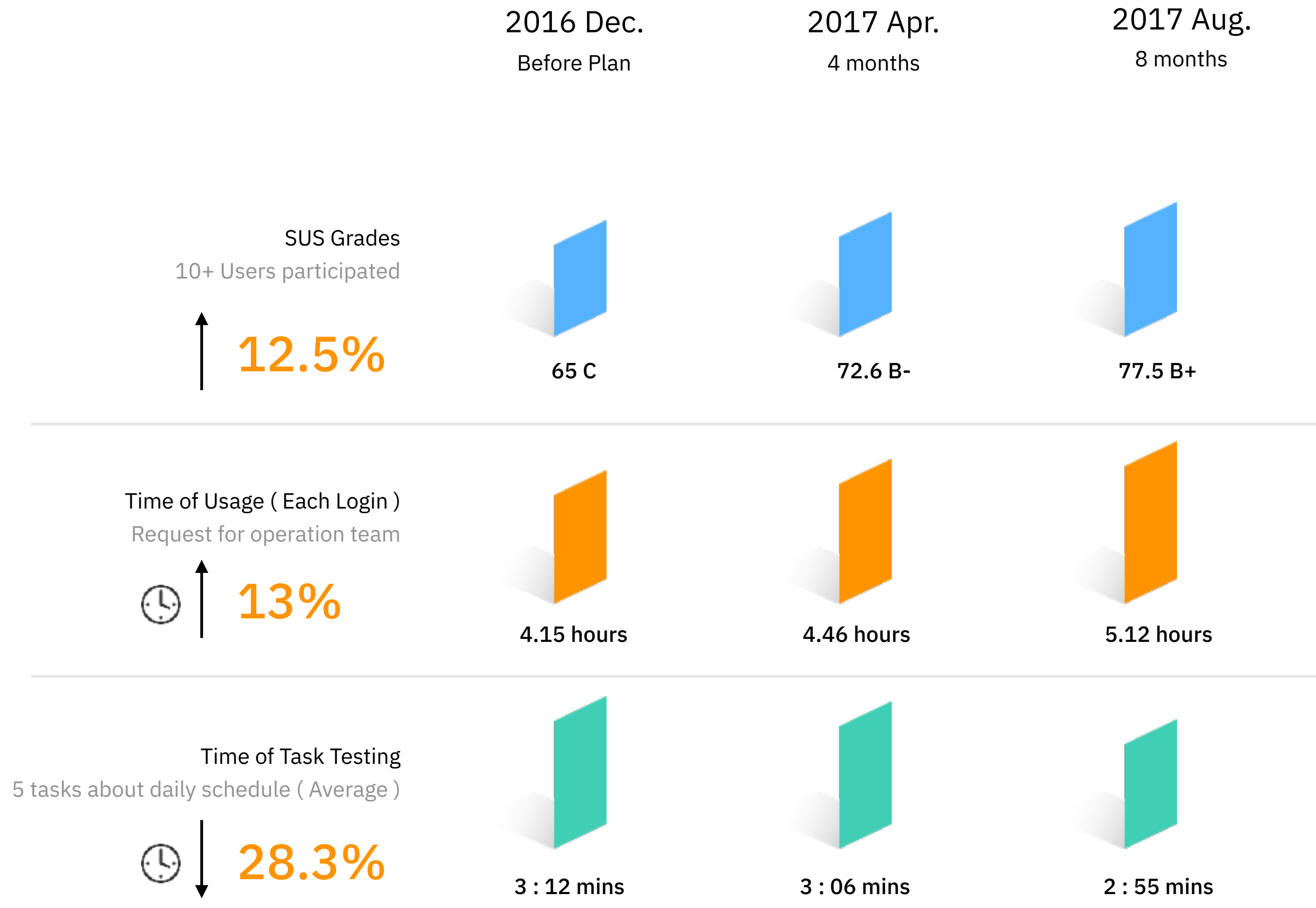
System Usability Scale (SUS) for Desktop

We designed the SUS questionnaire to evaluate our improvement result. We invite more than 10 users to fill out the form. Before we launched the plan on 2016 Dec., the grade is 65 (C). After we started to improve about 4 months, the grade is 72.6 (B-), after 4 months on 2017 Aug. the grade is up to 77.5 (B+).

SUS分数等级	评级	百分等级
84.1-100	A+	96-100
80.8-84	A	90-95
78.9-80.7	A-	85-89
77.2-78.8	B+	80-84
74.1-77.1	B	70-79
72.6-74	B-	65-69
71.1-72.5	C+	60-64
65-71	C	41-59
62.7-64.9	C-	35-40
51.7-62.6	D	15-34
0-51.7	F	0-14

SUS Grades Chart

The Outcomes of User Testing



Mobile Design

Mobile include two version in this APP, patients could use APP to book the calendar and get the feedback after treatment. Doctors and employees in clinic can review their own EMR, especially receptionist are more frequently use this APP to book appointment.



Desktop Design

Desktop is responsible for all of management in clinic, surrounding EMR, booking the treatment, patient management and cashier. The most biggest problem in our early research is booking treatment. In this iteration, the receptionist can define the color for specific patient. It's very helpful to them easily and quickly identify every status of patient.

